

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for PROLOG with cut and prove them equivalent in the Rocq prover. The first semantics closely models ELPI’s implementation: it is based on a stack of alternatives and is well suited for studying optimizations employed by PROLOG abstract machines. The second semantics is instead based on And-Or trees, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics to (i) describe introduction and elimination rules for disjunction in the presence of cut, which (as expected) depart from those of classical logic, and (ii) reason about the determinacy analysis introduced in ELPI version 3.0, which statically identifies computations that produce at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [19, 20, 9, 14]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [15, 3, 4, 26, 10, 11]. Other remarkable libraries include the Hierarchy-Builder library-structuring tool [5] and Derive [24, 25, 13], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI is equipped with a static analysis for determinacy [12] to help users control backtracking. ROCQ users are familiar with functional programming, but not necessarily with logic programming; in particular, uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checker allows the user to assert which predicates should be considered deterministic and then checks that these predicates behave like functions, i.e., they commit to their first solution and leave no *choice points* (places where backtracking could resume). A choice point correspond to an suspended *alternative* in the execution trace. In the following we use *choice point* and *alternative* as synonyms.

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [12]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [21] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations



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```

func f int -> int.           % f is a deterministic predicate. The two rules
f 1 2.                       % have non-verlapping heads, i.e. 1 != 2
f 2 3.
pred r int -> int.          % r is a non-deterministic predicate. The heads
r 0 0.                       % overlap: the query for 0 has three applicable
r 0 1.                       % rules.
r 0 2 :- !.
func g int -> int.          % g is deterministic: if the first rule succeeds, the
g A C :- r A B, f B C, !.    % second one is cut away, as well as the choice
g D F :- f D E, f E F.      % points left by the call to r.

```

■ **Figure 1** Running example of a ELPI program

used by standard PROLOG abstract machines [27, 1], but it makes reasoning about the scope of cut difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by cut are explicit and we prove the two semantics equivalent. Using the tree-based semantics, we then prove some properties about the impact of evaluating the cut in disjunctive queries. For example, unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 .

Finally, we use the formal semantics to prove the correctness of a static determinacy analysis [12]. Intuitively, nondeterminism arises when a computation can be completed by applying two different rules. We exclude this situation in two ways. Firstly, at most one rule should be used on a given query. This condition is satisfied if (i) the rules have non-overlapping heads: if two heads do not overlap in the inputs they accept, then no query can unify with both, or if (ii) we account for the presence of cut in rule premises: once a rule whose premises contain a cut succeeds, all remaining rules are discarded. Secondly, each rule of a deterministic predicate should (i) either call functions, this ensure that no choice point is left after the execution of the premises of the chosen rule, (ii) or call relations provided that they are followed by the cut operator, so that any allocated choice points preceeding the cut are discarded.

We show that if every rule defining a deterministic predicate passes this analysis, then any call to that predicate leaves no choice points.

2 Preliminaries: syntax and informal semantics of cut

In the entire paper we use Figure 1 as our running example, and we start by describing how we represent a program like that one. In the concrete syntax, variables are written with capital letters, and predicate application follows the λ -calculus style (no parentheses).

The description of a program is given by the record $\mathbb{P}$ and is shared by both semantics. A program consists of a list of rules R together with a signature environment sigT . We delay the discussion of signatures to Section 6 where we describe the static analysis that concerns them.

```

Inductive Tm := Symb of S | Pred of P | Var of V | App of Tm & Tm.
Inductive Atom := cut | call of Tm.
Record R := mkR { head : Tm; premises : list Atom }.
Record P := { rules : seq R; sig : sigT }.
Definition Σ := {fmap V -> Tm}.

```

■ **Figure 2** Definition of term, rule, program and substitution

A rule consists of a head term $\mathbf{Tm}$ and a list of premises. Each premise is an **Atom**, i.e. either the cut operator or a predicate call. Terms $\mathbf{Tm}$ include predicate symbols, data symbols, and variables. The syntax tree allows variables to be applied and predicates to be mixed with data. These higher order features play no role in the current paper that focuses on first order logic programming, but allows future extensions to full λ Prolog to be based on the same mechanization.

The set of variables $\mathbf{V}$, predicates $\mathbf{P}$ and data symbols $\mathbf{S}$ are countable. We represent substitutions Σ as finitely supported maps from variables to terms, using the finmap library [18]. The empty substitution is written ε .

The backchaining operation applies all rules to a query term; it is central to both semantics.

Definition `backchain` : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \mathbf{Tm} \rightarrow \Sigma \rightarrow \mathcal{F}_v * \text{list } (\Sigma * \text{list Atom})$

Since a rule may be applied multiple times, its variables must be freshened before each application. Accordingly, the `backchain` function takes a program, the set of variable names used so far ($\mathcal{F}_v$), a query term, and the current substitution. It returns an updated set of variable names together with the list of alternatives, i.e. the rules that apply. More precisely, for each applicable rule it returns the rule premises together with an updated substitution obtained by unifying the rule head with the query term.

In our mechanization, we abstract over the unifier and its properties. The function `unify` takes two terms t_1 and t_2 , together with a substitution σ , as input, and optionally returns a substitution σ' that extends σ . The expected behavior of `unify` is the standard one (see for example [8, section 2.5]), limited to the first order. Therefore, if t_1 and t_2 unify under σ , then $\sigma' t_1 = \sigma' t_2$, where σt denotes substitution application and $=$ denotes syntactic equality.

2.1 The cut operator

There are quite a few variants of the cut operator. The one we are interested in, i.e. the one used in ELPI, is known as *hard cut* (see, e.g., the documentation of SWI-PROLOG [23]). Whenever the `backchain` function returns a non-empty list of alternatives, the first one is explored immediately, while the others are suspended as choice points. Within a given alternative, premises are processed from left to right, executing each atom in turn. When a cut is encountered, it discards (i) all choice points created by `backchain` for the current call and (ii) all choice points created while executing the premises to the left of the cut.

For example, consider the program in Figure 1 and the query `g 0 R`. Both rules for `g` apply to this query. Backchaining therefore returns the following two alternatives:

$[(\sigma_1, [\mathbf{r} \ \mathbf{A} \ \mathbf{B}, \ \mathbf{f} \ \mathbf{B} \ \mathbf{C}, \ !]), (\sigma_2, [\mathbf{f} \ \mathbf{D} \ \mathbf{E}, \ \mathbf{f} \ \mathbf{E} \ \mathbf{F}])]$

with $\sigma_1 := \{\mathbf{A} \mapsto 0, \mathbf{C} \mapsto \mathbf{R}\}$ and $\sigma_2 := \{\mathbf{D} \mapsto 0, \mathbf{F} \mapsto \mathbf{R}\}$. For simplicity, we choose an example where variable refreshing plays no role, so we do not discuss the set $\mathcal{F}_v$ any further.

Execution continues with the first premise of the first alternative, namely `r A B` under σ_1 , i.e. `r 0 B`. The remaining alternatives are stored as choice points. Backchaining `r 0 B` returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [!])]$ where $\sigma_3 := \sigma_1 + \{\mathbf{B} \mapsto 0\}$; $\sigma_4 := \sigma_1 + \{\mathbf{B} \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{\mathbf{B} \mapsto 2\}$.

The next step of interpretation continues with `f B C` under σ_3 , that is, `f 0 R`, and leaves $[(\sigma_4, []), (\sigma_5, [!])]$ as new choice points. This time, backchaining fails (no rule for `f` matches input 0). We therefore backtrack to the most recently introduced choice point and retry `f B C` under σ_4 , i.e. `f 1 R`. This succeeds with $\sigma_6 := \sigma_4 + \{\mathbf{R} \mapsto 2\}$.

```

Inductive tree :=
  | KO | OK                                (* failure and success *)
  | Todo of Atom                          (* unexplored branch *)
  | Or  of option tree &  $\Sigma$  & tree      (*  $A \vee B_\sigma$  *)
  | And of tree & list Atom & tree.      (*  $A \wedge_{B_0} B$  *)

```

■ **Figure 3** Definition of And-Or tree

117 We now reach the cut. At this point there are still two unexplored alternatives: one from
 118 the third rule for **r** and one from the second rule for **g** (respectively, σ_5 and σ_2). Both are
 119 discarded, because they were created when, or after, the first rule for **g** was selected. In
 120 contrast, a cut does not discard choice points created earlier; in this example, there are none.
 121 The final solution is: $\{A \mapsto 0, B \mapsto 1, C \mapsto R, R \mapsto 2\}$.

122 In Sections 3 and 4, we provide fully mechanized operational semantics for PROLOG
 123 with cut. They differ in how they represent the ongoing computation, in particular how
 124 alternatives are stored and how they are pruned by cut.

125 *Notational conventions* In the following section we note $\mathbb{B}$ for the boolean type, and $\top$
 126 and $\perp$ for **true** and **false**. The constructors of the option type are written $\cdot t$ for **Some** t
 127 and $\square$ for **None**. Following the Mathematical Components library, on which we depend,
 128 we use $x.1$ and $x.2$ to denote the first and second projection applied to the pair x . List
 129 concatenation is the infix $++$, while list comprehension (i.e. mapping a function on a list) is
 130 written $[\text{seq } f \ x \mid x \leftarrow l]$.

131 3 And-Or Tree semantics

132 An And-Or tree is a stratified, variable-width tree, defined in Rocq as in Section 3. It
 133 consists of two kinds of layers that alternate: a disjunctive layer is followed by a conjunctive
 134 layer and vice versa. At the root (a query), the tree records a list of alternatives (an Or-layer);
 135 for each alternative, it records a list of subqueries to be solved (an And-layer). Intuitively,
 136 solving a query requires solving one alternative in the Or-layer, but all subqueries in the
 137 corresponding And-layer.

138 The tree is built incrementally. The **Todo** node represents an unexplored branch (an **Atom**).
 139 When the atom is a **call**, we use the backchaining procedure from Section 2 to compute the
 140 two layers to attach to the tree: alternatives form the Or-layer, and the premises of each
 141 applicable rule form the corresponding And-layer. This expansion step is performed by the
 142 **step** procedure described in Section 3.2.

143 In addition to the **Todo** node we have two distinguished nodes, **KO** and **OK**, which represent
 144 respectively the smallest failed and successful tree.

145 The **Or** node, written $A \vee B_\sigma$, represents the addition of an alternative B_σ to an existing
 146 disjunction A . The left subtree A is optional and is absent once that branch has been fully
 147 explored. The right alternative is paired with a substitution σ , which becomes the current
 148 substitution when backtracking to B .

149 The **And** node, written $A \wedge_{B_0} B$, represents the addition of a subquery B to the first
 150 choice point in A . We call B_0 the *reset point* for B : it represents the continuation for all
 151 alternatives of A except the first. That is, when backtracking occurs within A , the subtree
 152 B is replaced by the atoms in B_0 . An example is shown in Section 3.3.

153 A graphical representation of a tree is shown in Figure 6b. For compactness, we depict

```

Inductive tag := Expanded | CutBrothers | Failed | Success.
Definition step :  $\mathbb{P} \rightarrow \mathcal{F} \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F} * \text{tag} * \text{tree})$ 
Definition prune :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree}$ 
Inductive runT :  $\mathbb{P} \rightarrow \mathcal{F} \rightarrow \Sigma \rightarrow \text{tree} \rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F} \rightarrow \text{Prop}$ 

```

■ **Figure 4** Signatures of the tree-based semantics

154 And and Or nodes as n -ary (rather than binary), with children associating to the right. The
 155 KO node acts as the terminating element of an Or list, while OK is the terminating element of
 156 an And list.

157 3.1 The tree-based semantics

158 The tree is constructed incrementally by two recursive functions, **step** and **prune**. Both
 159 traverse the tree depth-first, left-to-right. The node to be explored in a tree is retrieved
 160 thanks to the **next_tree** (in Figure 5). When this node is OK we say the entire tree is a
 161 *success*, when it is KO we say the tree *failed*, otherwise the tree is *incomplete*. In Figure 6 we
 162 mark with a black arrow the result of **next_tree**.

163 Since all functions must terminate in Rocq, we define their transitive closure of **step**
 164 and **prune** as the inductive relation **runT**. More precisely **runT** iterates calls to **step** on
 165 incomplete trees to expand **Todo** nodes. When a tree fails it calls **prune** to find the next
 166 alternative and continues from there, if one exists.

167 3.2 The step procedure

168 The **step** procedure takes as input a program, a set of variables, a substitution, and a tree,
 169 and returns an updated set of variables, a **tag**, and an updated tree.

170 The set of variables is assumed to contain all variables occurring in the tree. It is passed
 171 to backchain so that rules can be refreshed correctly.

172 The **tag** (see Figure 4) indicates which kind of step was performed. **Failure** and **Success**
 173 are returned for trees that satisfy, respectively, the **failed** and **success** tests. **CutBrothers**
 174 indicates the interpretation of a superficial cut: **step** was called on an And-layer and its

¹ The substitutions $\sigma_1 \dots \sigma_6$ are from Section 2.1. R_1, R_2 , and R_3 are reset points and correspond respectively to the lists of atoms $[f \ Y \ Z, !]$, $[!]$, and $[f \ Y \ Z]$.

```

Fixpoint next  $\sigma \ t : \Sigma * \text{tree} :=$ 
  match  $t$  with
  | Todo _ | KO | OK  $\Rightarrow (\sigma, t)$ 
  | Or  $\square \ \sigma' \ B \Rightarrow \text{next } \sigma' \ B$ 
  | Or  $\cdot A \ \_ \_ \Rightarrow \text{next } \sigma \ A$ 
  | And  $A \ \_ \ B \Rightarrow$ 
    let  $(\sigma', pA) := \text{next } \sigma \ A$  in
    if  $pA == \text{OK}$  then  $\text{next } \sigma' \ B$ 
    else  $(\sigma', pA)$ 
  end.

Definition next_subst  $\sigma \ t := (\text{next } \sigma \ t).1.$ 
Definition next_tree  $t := (\text{next } \varepsilon \ t).2.$ 

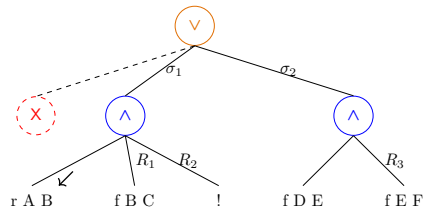
Definition success  $t := \text{next\_tree } t == \text{OK}.$ 
Definition failed  $t := \text{next\_tree } t == \text{KO}.$ 
Definition incomplete  $t :=$ 
  if  $\text{next\_tree } t$  is Todo _ then  $\top$  else  $\perp$ .

```

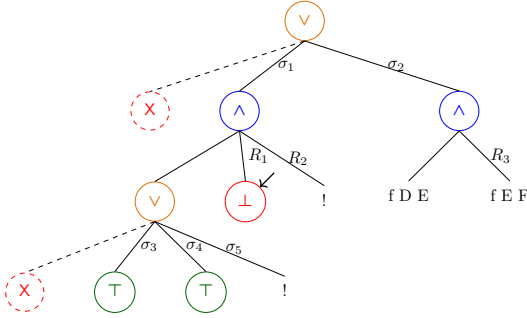
■ **Figure 5** next and derived functions

g 0 R

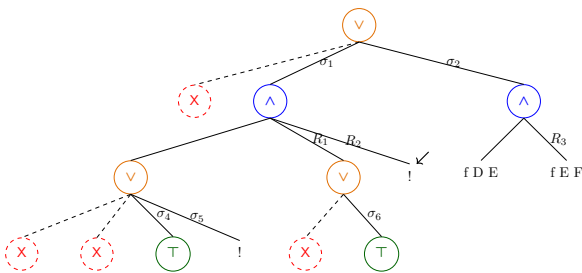
(a) Initial query



(b) Tree after step on g 0 R

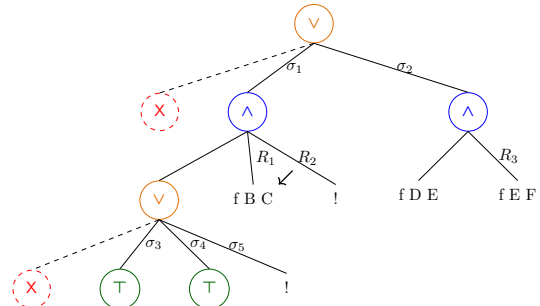


(d.1) step on Figure 6c

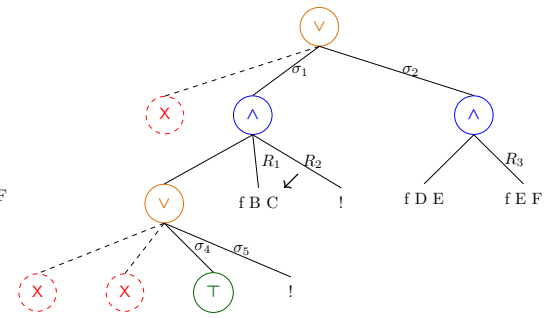


(e) step on Figure 6d.2

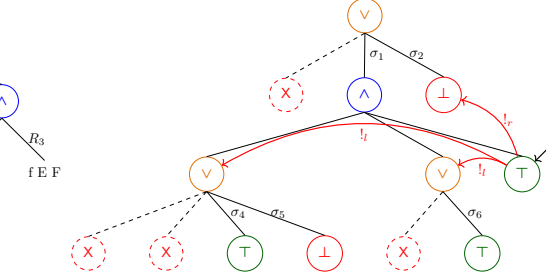
Figure 6 Some tree representations¹



(c) step on Figure 6b



(d.2) prune on Figure 6d.1



(f) step on Figure 6e

175 **next_tree** is the cut atom. Finally, **Expanded** accounts for the interpretation of predicate
 176 calls and non-superficial cuts.

177 The **step** procedure evaluates atoms. In particular, it leaves the tree unchanged when
 178 the tree is *success* or *failed*.

179 ► **Lemma 1** (*succ_step_iff*). $\forall p \ v \ \sigma \ t, \text{ success } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Success}, t).$

180 ► **Lemma 2** (*fail_step_iff*). $\forall p \ v \ \sigma \ t, \text{ failed } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Failed}, t).$

181 The **step** procedure expands **Todo** nodes by calling **backchain**, which returns a list l of
 182 alternatives. If l is empty, then the current call atom is replaced by **K0**. Otherwise, it is
 183 replaced by a right-skewed tree of **Or** nodes, where each leaf corresponds to a list of atoms
 184 $e \in l$. If e is empty, the leaf is **OK**; otherwise, it is a right-skewed tree of **And** nodes.

185 Figure 6 depicts the tree corresponding to the running example (Section 2.1) as it
 186 undergoes calls to **step**. We run query $q \ 0 \ R$ against the program P in Figure 1. Let c_0 ,
 187 c_1 , and c_2 be the children of the root. By construction, c_0 is the $\square$ tree, while c_1 and c_2
 188 correspond respectively to the premises of rules r_1 and r_2 in P . Note that c_1 (resp. c_2) is
 189 associated with substitution σ_1 (resp. σ_2), whose values are the same as in Section 2.1. Reset
 190 points are defines as follows: $R_1 := [\mathbf{f} \ Y \ Z, !]$, $R_2 := [!]$, and $R_3 := \mathbf{f} \ Y \ Z$. Intuitively, they
 191 record the lists of atoms used to generate the corresponding subtree. Other examples of
 192 **step** performing a tree expansion via **backchain** appear in Figures 6c–6e.

193 3.2.1 The interpretation of a cut

194 The step corresponding to a cut performs two actions: (i) the cut node is replaced by **OK**; and
 195 (ii) the subtrees in the scope of **Cut** are pruned. More precisely, (ii) prunes the left siblings
 196 of the **Cut** node (in the same **And**-layer, denoted $!_l$) and prunes the right uncles of **Cut** (in
 197 the one-level-up **Or**-layer, denoted $!_r$).

198 An example of the impact of cut is shown in Figure 6f. The red arrows indicate the
 199 scope of the cut and are labeled with $!_r$ and $!_l$ to indicate which pruning operation is applied
 200 to each pointed subtree. Note that the evaluation of a cut keeps the path leading to the
 201 cut node unchanged, in the sense that substitution σ_4 is preserved, while the path under
 202 substitution σ_5 (which also succeeds) is replaced by **K0**. This amounts to discarding the third
 203 rule of predicate **r**.

204 3.3 The prune procedure

205 When **backchain** returns an empty list, **step** produces a tree that *failed* and the computation
 206 must backtrack. The **prune** procedure is responsible for producing the new tree from which
 207 the computation can continue.

208 In Figure 6d.1 we encounter a failure because no rule applies to the atom $\mathbf{f} \ B \ C$ under
 209 substitution σ_3 , which maps **B** to 0. The **prune** procedure prunes the path leading to this
 210 failure and resets the current sub-query to its original shape. More precisely, it kills the **OK**
 211 node under σ_3 (since that branch has been fully explored and produced no solution), and
 212 then replaces the **K0** node with the information stored in the reset point R_1 . This restores
 213 the call $\mathbf{f} \ B \ C$, which can then be reconsidered under substitution σ_4 .

214 Furthermore, **prune** serves another important purpose. Its behavior depends on the
 215 boolean flag it receives as input: **prune** $\perp t$ recursively removes all failures (if any) from t ,
 216 whereas **prune** $\top t$ removes the first success in t . For example, the tree in Figure 6f contains
 217 a successful path that maps **R** to 2. Calling **prune** on this tree returns $\square$, since there are no
 218 alternatives in the tree after the success.

$$\begin{array}{c}
\frac{\text{success } t \quad \text{next_subst } \sigma \ t = \sigma' \quad \text{prune } \top \ t = t'}{\text{runT } p \ v \ \sigma \ t \cdot(\sigma', t') \perp v} \text{StopT} \\
\\
\frac{\text{incomplete } t \quad b' = (\text{tg} = \text{CutBrothers}) \vee b \quad \text{step } p \ v \ \sigma \ t = (v', \text{tg}, t') \quad \text{runT } p \ v' \ \sigma \ t' \ r \ b \ v''}{\text{runT } p \ v \ \sigma \ t \ r \ b' \ v''} \text{StepT} \\
\\
\frac{\text{failed } t \quad \text{prune } \perp \ t = \cdot t' \quad \text{runT } p \ v \ \sigma \ t' \ r \ n \ v'}{\text{runT } p \ v \ \sigma \ t \ r \ n \ v'} \text{BackT} \\
\\
\frac{\text{prune } \perp \ t = \square}{\text{runT } p \ v \ \sigma \ t \ \square \perp v} \text{FailT}
\end{array}$$

■ **Figure 7** Rule system for `runT`

Some interesting properties of `prune` are shown below; they illustrate how `prune` complements `step` with respect to failed, successful, and incomplete trees.

► **Lemma 3** (`incomplete_prune_id`). $\forall b \ t, \text{incomplete } t \rightarrow \text{prune } b \ t = \cdot t$.

► **Lemma 4** (`prune_Some`). $\forall b \ t \ t', \text{prune } b \ t = \cdot t' \rightarrow \text{failed } t' = \perp$.

► **Lemma 5** (`prune_None`). $\forall t, \text{prune } \perp \ t = \square \rightarrow \text{failed } t$.

3.4 The `runT` inductive

The inductive relation `runT` relates four inputs to three outputs. The inputs are a program, a set of variables, an initial substitution σ , and a tree t . The outputs are an optional result r , a boolean b , and an updated set of variable names.

The main result r is $\square$ when the execution fails; otherwise when $r = \cdot(\sigma', t')$ the execution succeeded and produced the solution σ' ; t' represents the remaining, not-yet-explored alternatives. The boolean b records whether a superficial cut was evaluated during execution.

The `runT` predicate has four cases, depicted as inference rules in Figure 7. (*StopT*) If `next_tree` t is a success, then the substitution σ' is extracted from t (starting from σ) and the input tree is cleaned of its successful path. (*StepT*) If `next_tree` t is an atom, then `step` is invoked to evaluate t , and `runT` is called recursively on the updated tree. (*BackT*) If `next_tree` t is a failure, then `prune` is called to clear the failed path; if the resulting cleaned tree exists, `runT` is called recursively on it. Finally, (*FailT*) accounts for trees with no solutions.

The set of rules defining `runT` is deterministic.

► **Lemma 6** (`runT_det`). $\forall p \ v_0 \ \sigma \ t \ r \ r' \ b \ b' \ v \ v', \text{runT } p \ v_0 \ \sigma \ t \ r \ b \ v \rightarrow \text{runT } p \ v_0 \ \sigma \ t \ r' \ b' \ v' \rightarrow r' = r \wedge v' = v \wedge b' = b$.

The proof is by induction on the first `runT` derivation and is immediate, because the rules are selected based on `next_tree` t . In particular, the hypotheses `success` t , `incomplete` t , and `failed` t are mutually exclusive. Moreover, by Lemma 5, the hypothesis of *FailT* implies that t failed; hence *FailT* is exclusive with *StopT* and *StepT*, and it is also exclusive with *BackT* since they impose different requirements on the output of `prune`.

The boolean output of `runT` allows us to state interesting properties about the `Or` node in the presence of cut. Here we report only a selection of those we proved.

► **Lemma 7** (`runSST_or`). $\forall p \ v \ v' \ \sigma \ \sigma' \ A \ A' \ \sigma_1 \ B, \text{runT } p \ v \ \sigma \ A \cdot(\sigma', \cdot A') \top v' \rightarrow$

$\text{runT } p \ v \ \sigma \ (\cdot A \vee B_{\sigma_1}) \cdot (\sigma', \cdot (\cdot A' \vee KO_{\sigma_1})) \perp v'.$

247 ▶ **Lemma 8** (runNT_or). $\forall p \ v \ v' \ \sigma \ A \ \sigma_1 \ B, \text{runT } p \ v \ \sigma \ A \ \square \top v' \rightarrow$
 $\text{runT } p \ v \ \sigma \ (\cdot A \vee B_{\sigma_1}) \ \square \perp v'.$

248 ▶ **Lemma 9** (runNF_or). $\forall p \ v_0 \ v_1 \ v_2 \ \sigma \ A \ \sigma_1 \ \sigma_2 \ B \ B' \ b,$
 $\text{runT } p \ v_0 \ \sigma \ A \ \square \perp v_1 \rightarrow \text{runT } p \ v_1 \ \sigma_1 \ B \cdot (\sigma_2, B') \ b \ v_2 \rightarrow$
 $\text{let } sR := \text{omap } (\text{fun } x \Rightarrow \square \vee x_{\sigma_1}) \ B' \text{ in}$
 $\text{runT } p \ v_0 \ \sigma \ (\cdot A \vee B_{\sigma_1}) \cdot (\sigma_2, sR) \perp v_2.$

249 Lemmas 7–9 correspond to the introduction rules for disjunction. If A executes a cut, one
 250 cannot obtain $A \vee B$ from B : the execution of A , whether it ultimately succeeds or fails, makes the
 251 success of B irrelevant (the cut discards it). In the first lemma, B is forced to be KO ; in the second
 252 one, the failure of A absorbs B . The last lemma is connected to the depth-first exploration of the
 253 tree: proving a disjunction from its right-hand side can only occur after the left-hand side has been
 254 disproved. Depicted as rules:

$$\frac{A \quad (A \text{ cuts})}{A \vee \top} (\vee_{il}) \quad \frac{\neg A \quad (A \text{ cuts})}{\neg(\neg A \vee B)} (\vee_{ir1}) \quad \frac{\neg A \quad B \quad (A \text{ does not cut})}{A \vee B} (\vee_{ir2})$$

255 Lemmas 10 and 11 are elimination rules for disjunction.

256 ▶ **Lemma 10** (run_orSST). $\forall p \ v \ v' \ \sigma \ \sigma' \ \sigma_1 \ A \ A' \ B \ B',$
 $\text{runT } p \ v \ \sigma \ (\cdot A \vee B_{\sigma_1}) \cdot (\sigma', \cdot (\cdot A' \vee B'_{\sigma_1})) \perp v' \rightarrow$
 $\exists b, \text{runT } p \ v \ \sigma \ A \cdot (\sigma', \cdot A') \ b \ v' \wedge B' = \text{if } b \text{ then } KO \text{ else } B.$

257 ▶ **Lemma 11** (run_orSNT). $\forall p \ v \ v' \ \sigma \ \sigma' \ \sigma_1 \ A \ B \ B',$
 $\text{runT } p \ v \ \sigma \ (\cdot A \vee B_{\sigma_1}) \cdot (\sigma', \cdot (\square \vee B'_{\sigma_1})) \perp v' \rightarrow$
 $(\exists b, \text{runT } p \ v \ \sigma \ A \cdot (\sigma', \square) \ b \ v' \wedge \text{if } b \text{ then } B' = KO \text{ else } \text{prune } \perp B = \cdot B') \vee$
 $(\exists v_2 \ b, \text{runT } p \ v \ \sigma \ A \ \square \perp v_2 \wedge \text{runT } p \ v_2 \ \sigma_1 \ B \cdot (\sigma', \cdot B') \ b \ v').$

258 From $A \vee B$, when A is not inert (i.e. not already explored), we cannot deduce A or B
 259 independently. Instead, we can only derive a weakened conclusion of the form $\top \vee B'$, where B'
 260 provides no information in the case where the exploration of A performs a cut. This yields an
 261 elimination rule that is stronger than the usual one (depicted on the left). If A does not cut, the
 262 elimination rule has the usual two premises, together with the additional constraint that whenever
 263 B holds, A must have failed. This again reflects the depth-first traversal of the proof tree.

$$\frac{A \vee B \quad \frac{[A] \quad C}{C} \quad (A \text{ cuts})}{C} (\vee_{e1}) \quad \frac{A \vee B \quad \frac{[A] \quad [\neg A, B] \quad C}{C} \quad (A \text{ does not cut})}{C} (\vee_{e2})$$

264 It is worth recalling that the “ A cuts” side-condition in the rules above is precisely the boolean
 265 result returned by runT . Without this information, it is impossible to formulate these rules: it is
 266 not merely a matter of whether A contains a syntactic occurrence of cut, but whether that cut is
 267 actually executed during the (dis)proof of A (in rules $(\vee_{ir1})$ and $(\vee_{ir2})$). Indeed, an atom preceding
 268 the cut may fail, so the cut is never reached.

269 As anticipated in the introduction these lemmas contradict the commutativity of disjunction.

4 Stack-based semantics

271 The stack-based semantics we present is very close to what is implemented by the ELPI interpreter.
 272 This semantics is similar to the one proposed by Pusch in [21], which is in turn closely related to
 273 the semantics given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized this semantics in
 274 Isabelle/HOL, together with several optimizations that are also present in the Warren Abstract
 275 Machine [27, 1].

$$\begin{array}{l}
\text{Definition } \text{save_gs } a \ g \ b := \\
\quad [\text{seq } (x, a) \mid x \leftarrow b] ++ g. \\
\text{Definition } \text{save_as } a \ g \ b := \\
\quad [\text{seq } (x.1, \text{save_gs } a \ g \ x.2) \mid x \leftarrow b]. \\
\text{Definition } \text{stepS } p \ v \ t \ \sigma \ a \ g := \\
\quad \text{let } (v', r) := \text{backchain } p \ v \ t \ \sigma \ \text{in} \\
\quad (v', \text{save_as } a \ g \ r). \\
\frac{}{\text{runS } p \ v \ ((\sigma, \emptyset) :: a) \cdot (\sigma, a)} \text{StopS} \quad \frac{\text{runS } p \ v \ ((\sigma, g) :: ca) \ r}{\text{runS } p \ v \ ((\sigma, (\text{cut}, ca) :: g) :: _) \ r} \text{CutS} \\
\frac{\text{stepS } p \ v \ t \ \sigma \ a \ g = (v_1, a') \quad \text{runS } p \ v_1 \ (a' ++ a) \ r}{\text{runS } p \ v \ ((\sigma, (\text{call } t, _) :: g) :: a) \ r} \text{CallS} \quad \frac{}{\text{runS } p \ _ \ \emptyset \ \square} \text{FailS}
\end{array}$$

■ **Figure 8** Rule system for runS

276 The execution state is a stack of alternatives. Each alternative is a list of goals. Intuitively it
 277 amounts to a disjunctive normal form: each stack item is a self contained computation, coupling a
 278 substitution with all the goals to be solved.

279 Goals pair an atom with a list of the *cut-to* alternatives, making the two datatypes mutually
 280 recursive, but these alternatives have a very different role. They have to be understood as pointers
 281 to lower levels of the stack itself. This datum is used to implement the cut, i.e. to prune the stack
 282 of alternatives.

Inductive alts := nilA | consA of (Σ * goals) & alts
with goals := nilG | consG of (Atom * alts) & goals

283 The stack-based semantics is described by the runS inductive predicates.

284 **Inductive** runS (p: $\mathbb{P}$) : $\mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop}$

285 The runS predicates relates a set of variables and a list of alternatives to optional result. $\square$ stands
 286 for failure, i.e., no solution, whereas $\cdot(\sigma, a)$ represent success with σ begin the solution and a as the
 287 list of non-yet-explored alternatives.

288 The rule system for runS is given in Figure 8 and is made by 4 cases. We distinguish two main
 289 cases. If we run out of alternatives we fail: (*FailS*) stops the execution and return $\square$. If the list of
 290 alternatives is $(\sigma_0, h) :: a$, then we look at the list of goals h (under the substitution σ_0). If h is
 291 empty, (*StopS*) stops the execution with a success σ_0 leaving a as the unexplored alternatives. If h
 292 is not empty we look at its head goal (t, ca) where t is the atom and ca is its cut-to alternatives. If t
 293 is a cut rule (*CutS*) continues to evaluate the tail of h under with alternatives ca . Finally, when
 294 t is a call, rule (*CallS*) backchains: for each rule it pairs each premise with the current stack of
 295 alternatives, and prepends the obtained goals to the tail of h . Note that if backchain returns the empty
 296 list, (*CallS*) second premise amounts to exploring the next item in the current stack.

297 4.1 Example of execution

298 We propose an example of the execution of runS in Figure 9. The arrows in the images represent
 299 the cut-to pointers. We have not drawn arrows coming out of call atoms since the interpreter
 300 ignores the cut-alternatives in this case. The evolution of the stack in Figure 9 mirrors the example
 301 in Section 2.1: we evaluate the first goal of the first alternative. During backchaining, we add a
 302 pointer to the remaining alternatives to each new cut operator. If a cut is evaluated, we replace the
 303 remaining alternatives with the cut-to alternatives. For example, in Figure 9f, evaluating the cut
 304 discards the second and third elements in the stack. Backtracking is performed by simply consuming
 305 the first alternative from the stack, as shown in Figure 9d.

4.2 Inadequacy of the stack-based semantics for formal reasoning

The lemmas that we presented in Section 3.4 are hard to express in the stack-based semantics, due to the lack of structure in the stack to delimit the scope of the cut operator. For example, consider a stack $a_0, a_1, \dots, a_n$. If a_0 succeeds but executes a cut, it is hard to relate the remaining alternatives $a_1, \dots, a_n$ to the alternatives $b_1, \dots, b_m$ that survive the cut, since the only simple invariant is that $b_1, \dots, b_m$ is a suffix of $a_1, \dots, a_n$. If $a_1, \dots, a_n$ were generated before the call whose execution generates the cut in a_0 , then they all stay; but if some were generated afterwards, they are discarded. Expressing this precisely would require attaching, at the very least, time-stamps (or depths) to goals, which amounts to a cumbersome encoding of a tree.

5 Equivalence of the two semantics

In order to relate the two semantics we have to relate the computations states, i.e., the And-Or tree and the stack of alternatives. We define a translation from trees to stacks, called **t2l**, but we do not explicitly provide the converse translation, an economy allowed by the proof technique we employ, inspired by [2].

Before describing the translation of trees to stacks (in Section 5.2) we need to define the notion of *valid tree*. The **tree** datatype indeed contains trees that cannot be generated by **runT** via **step** or **prune**, for example all the trees that do not correspond to a depth-first left-to-right tree construction strategy.

5.1 Valid trees

The class of valid trees is captured by the **valid_tree** function shown in Figure 10. While all base cases of **tree** are valid, we need to analyze the **Or** and **And** constructors more carefully.

For the **Or** constructor, we distinguish two cases depending on whether the left-hand side is present. If it is not, then the right-hand side must be a valid tree. Otherwise, the left-hand side must be a valid tree, and the right-hand side must either be the **K0** tree or be unexplored. The former case occurs when the right-hand side is cut away by the evaluation of a superficial cut in the left-hand side. In the latter case we use the **base_or** predicate to check that **B** is the right-skewed tree formed by a disjunction of conjunctions, generated after a successful backchain for a **call**.

For the **And** constructor, the left hand side is required to be a valid tree. The shape of the right hand side depends on whether the left hand side is a success. If it is not successful, then the right hand side must not be explored, i.e. **B** must be the right-skewed tree containing conjunctions of premise atoms. This is enforced using the **big_and** procedure, which generates a tree from the reset point B_0 (the same procedure used to transform each alternative output by **backchain** into the

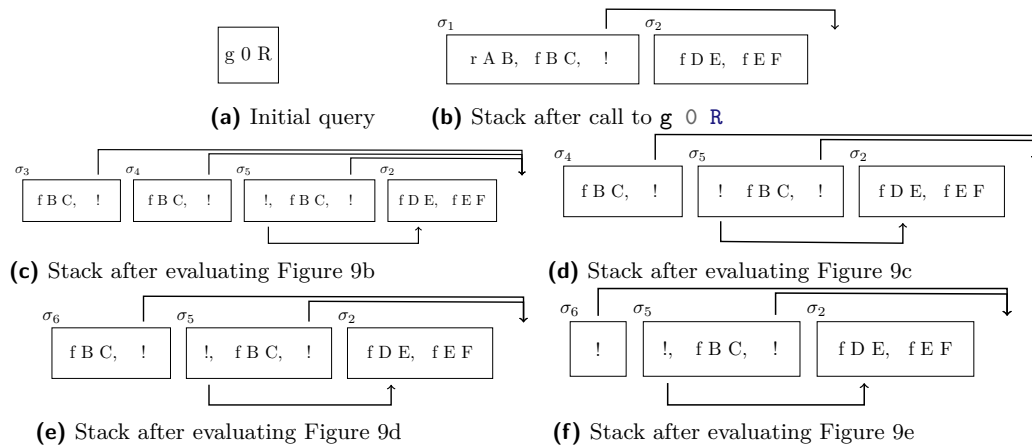


Figure 9 Example of execution

```

Fixpoint valid_tree A :=
  match A with
  | Todo _ | OK | KO => T
  | Or □ _ B => valid_tree B
  | Or ·A _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

Fixpoint t2l t σ (cp: alts) : alts :=
  match t with
  | OK      => [(σ, ∅)]
  | KO      => ∅
  | TA a    => [(σ, [(a, ∅)])]
  ...

```

■ Figure 10 Valid tree and Tree to list

338 And-layer of the resulting tree). When the left hand side is successful, then the right hand side may
 339 have been explored, and consequently must be a valid tree.

340 5.2 From trees to stacks

341 The `t2l` recursive function translates a tree to a stack. For space constraints Figure 10 only gives
 342 the base cases, while we describe the recursive cases in the following text.

343 The function `t2l` takes as input the tree t_0 , a substitution σ_0 , and a list of choice points cp .
 344 These choice points represent alternatives that appear at the same level of the current tree, such as,
 345 for example, the right siblings of an `Or` node. The substitution is used to determine under which
 346 substitution the subtree being computed lives.

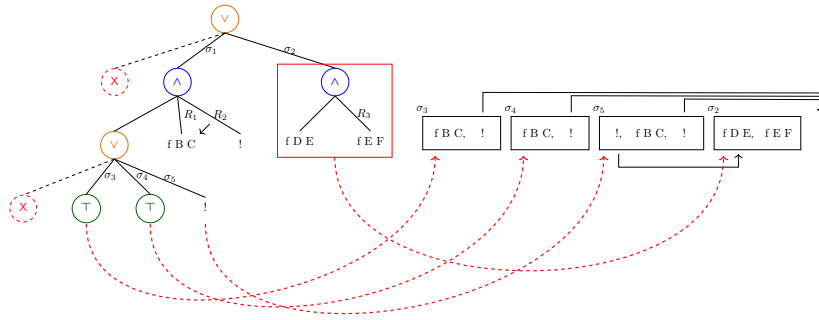
347 The `OK` node represents one successful alternative; therefore it is translated into a singleton
 348 list whose only element has an empty list of goals. The `KO` node represents failure, hence it is
 349 mapped to the empty list of alternatives. Finally, atoms are mapped to a singleton list containing
 350 one alternative whose only goal is that atom, with an empty cut-to list. At this stage, the cut-to
 351 list cannot point to cp : atoms are not right-uncle subtrees, whereas cp represents alternatives that
 352 will actually be discarded if the atom turns out to be a cut. The correct cut-to list is therefore
 353 determined later, once the trees corresponding to sibling subtrees have been inspected.

354 The translation of $A \vee B_\sigma$ creates two list of alternatives, one for the A and the other for B .
 355 The alternatives of B are valid choice points for A and should be passed to the translation of A .
 356 The two lists of alternatives can then be concatenated. Moreover, every atom occurring one level
 357 below cp (i.e. in A or B) should add cp as its cut-to alternatives.

358 The translation of $A \wedge_{B_0} B$ starts with the translation of A . If the translation of A is an empty list
 359 we return the empty list of alternatives without even touching B nor B_0 , since an empty translation
 360 for A indicates failure and this in turn implies the failure of the entire conjunction. When the
 361 translation of A is a list $(\sigma_0, g) :: a$, the B node represents the continuation of the first alternative g ,
 362 whereas B_0 is the continuation for each alternative in a .

363 5.2.1 Example of `t2l`

364 The translations of the trees in Figure 6 are shown in Figure 9. The letters in the figures relate each
 365 tree to the corresponding stack. For example, 6b is translated into 9b. We also note that 6d.1 and
 366 6d.2 are both translated into 9d, showing that `t2l` is not injective: different trees can map to the
 367 same stack. Consequently, there are transitions in `runT` that are not visible in `runS`.



368

369 As an example of a call to `t2l`, we take the tree and the stack above, which are respectively taken
 370 from Figures 6c and 9c. The red arrows point from the head of an alternative in the tree to the head
 371 of the corresponding cell in the stack. We focus on the deepest `Or` node in the tree. This subtree
 372 is a disjunction with four children. The first is ignored, since it is $\square$. The success node below σ_3
 373 has `f B C` and the cut operator as continuation, corresponding to the first cell in the stack. We
 374 note that the cut operator points outside the stack, since the scope of this cut eliminates its right
 375 uncle. The success node below σ_4 has R_1 as continuation, where R_1 is the list of goals `f B C, !`;
 376 this is translated into the second cell of the stack. Finally, the cut operator under σ_5 also has R_1 as
 377 continuation. It corresponds to the third alternative. We can see that the first cut points to the
 378 translation of the subtree under σ_2 , since it is one `Or`-layer above its right uncles.

379 5.3 Equivalence proof

380 The tree-based semantics exposes more information than needed, hence we hide it behind existentials.

381 ► **Definition 12** ($\text{runT}' p v \sigma t r$). $\exists b v', \text{runT } p v \sigma t r b v'$.

382 From Figures 6 and 9, we observe that, upon backtracking, `runT` may perform more steps than
 383 `runS` before returning a solution or a failure. These silent transitions must be skipped when proving
 384 the equivalence between the two semantics. Therefore, we prove the following lemma.

385 ► **Lemma 13** (pruneF_t2l). $\forall t t' b, \text{valid_tree } t \rightarrow \text{failed } t \rightarrow \text{prune } b t = \cdot t' \rightarrow$
 $\forall l \sigma, t2l t \sigma l = t2l t' \sigma l$.

386 **Proof.** By induction on the tree t . ◀

387 The first direction of the equivalence states that the execution of a valid tree is matched by
 388 the execution of the stack obtained by translating the tree. Moreover, the unexplored alternatives
 389 returned as a tree correspond to those returned as a stack.

390 ► **Theorem 14** (tree_to_elpi). $\forall p t \sigma r, \text{let } v := \text{vars_tree } t \setminus \text{vars_sigma } \sigma \text{ in}$

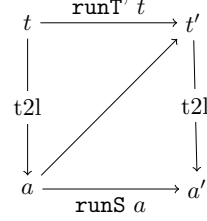
$\text{valid_tree } t \rightarrow$
 $\text{runT}' p v \sigma t r \rightarrow$
 $\text{let } a := t2l t \sigma \emptyset \text{ in}$
 $\text{let } r' := \text{if } r \text{ is } \cdot(\sigma', t) \text{ then } \cdot(\sigma', t2l (\text{odflt } KO t) \sigma \emptyset)$
 $\text{else } \square \text{ in}$
 $\text{runS } p v a r'$.

391 **Proof.** We proceed by induction on the execution of `runT'`. There are four cases to consider. The
 392 first case arises from `StopT`; it deals with successful trees. A successful tree must start with an empty
 393 list, and therefore we conclude using `StopS`. The case for `StepT` requires treating two subcases: `step`
 394 evaluates either a cut or a call. Accordingly, we apply `CutS` or `CallS`, followed by the induction
 395 hypothesis. The case for `BackT` is silent: it represents the non-injective behavior of `t2l`; however,
 396 thanks to Lemma 13 and the induction hypothesis, the conclusion is proved. The last case arises
 397 from the derivation `FailT`. It is easily proved using `FailS` and the fact that if `prune` returns $\square$ when
 398 trying to erase failure in t , then `t2l` applied to t returns the empty list. ◀

399 The converse direction is stated following [2]: instead of using a function to translate a stack
 400 into a tree it states that any tree that translates to the given stack has an equivalent execution, i.e.
 401 gives the same solution and returns a unexplored tree that translates to the unexplored stack.

402 ► **Theorem 15** (*elpi_to_tree*). $\forall p \ v \ a \ r, \text{runS } p \ v \ a \ r \rightarrow$
 $\forall \sigma \ t, \text{valid_tree } t \rightarrow t2l \ t \ \sigma \ \emptyset = a \rightarrow$
 $\text{if } r \text{ is } (\sigma', a') \text{ then}$
 $\exists t', \text{runT}' p \ v \ \sigma \ t \cdot (\sigma', t') \wedge t2l \ (\text{odflt } KD \ t') \ \sigma \ \emptyset = a'$
 $\text{else runT}' p \ v \ \sigma \ t \square.$

403 **Proof.** The proof is based on the idea explained in [2, Section 3.5.1] and
 depicted in the figure on the right: we have a tree t whose translation is
 a , we proceed by induction on the execution of **runS**. This gives us not
 only the output a' but that hypothesis that it exists a tree t' such that its
 translation to a stack is a' . ◀



404 Stating the converse direction in a symmetric way would require a function **12t** transforming a
 405 stack a into a tree t . Writing this function is far from trivial, since the structure of the tree is lost by
 406 **save_as** and **runT** that intuitively keep the state as a big disjunction of conjunctions by distributing
 407 conjunctions over disjunctions. Moreover, one would need to define a notion of valid stack that is
 408 equally intricate.

409 The constructive nature of the logic of Rocq tells us that the proof above provides an effective
 410 function translating stacks into trees. However, that function also uses an execution trace of the
 411 stack-based semantics, from which it recovers the missing tree structure.

412 6 Case study: determinacy analysis

413 An interesting application of the tree semantics is determinacy checking, introduced in version 3 of
 414 the ELPI language [12]. The checker takes as input the signatures of the predicates declared by the
 415 user and verifies that the rules of a deterministic predicate generate at most one solution; we call
 416 this particular kind of predicate a function. Thanks to this static verification, the user can better
 417 control unexpected backtracking.

418 A predicate behaves deterministically if two conditions are satisfied: *mutual exclusion* guarantees
 419 that at most one rule can be successfully applied to any given query, whereas *rule checking* ensures
 420 that each rule does not create choice points, for example by calling non-deterministic predicates.

421 Determinacy checking is strongly related to the modes of predicates: we differentiate between
 422 input and output arguments. A deterministic call relates to its inputs only one output. In the
 423 literature, we find several works on determinacy checking, namely [22, 17, 7]. These works need
 424 a mode checker that statically ensures that, in a query with ground inputs, each recursive call is
 425 performed with ground inputs.

426 In our mechanization, and in particular in the ELPI language, we adopt a special unification
 427 routine for input arguments, called **matching** [1, 28]. Similarly to **unify**, **matching** takes the query
 428 term t_1 and the pattern t_2 and returns an optional substitution σ' that does not assign any variable
 429 in the query.

430 The function **matching** has the following property:

431 ► **Axiom 1** (*matching_disj*). $\forall \sigma \ \sigma' \ t_1 \ t_2, \text{vars } t_1 \# \text{domf } \sigma \rightarrow \text{vars } t_1 \# \text{vars } t_2 \rightarrow$
 $\text{matching } t_1 \ t_2 \ \sigma = \cdot \sigma' \rightarrow \exists e, \text{domf } \sigma' = \text{domf } \sigma \cup e \wedge e \subseteq \text{vars } t_2.$

432 If the sets of variables occurring in the two terms are disjoint, and the variables in the support
 433 of σ are disjoint from the variables in t_1 , then when matching succeeds, only the variables in t_2 are
 434 assigned; that is, $t_1 = \sigma' t_2$, and the variables in t_1 are not assigned in σ' .

435 The **backchain** function uses **matching** or **unify** on the arguments q depending if the argument
 436 is in *input* or *output* mode. We assume two important properties of our unification routines.

437 ► **Axiom 2** (`unif_trans`). $\forall t_1 t_2 t_3 \sigma, \text{unify } t_1 t_2 \sigma \rightarrow \text{unify } t_2 t_3 \sigma \rightarrow \text{unify } t_1 t_3 \sigma.$

438 ► **Axiom 3** (`match_unif`). $\forall t_1 t_2 \sigma, \text{matching } t_1 t_2 \sigma \rightarrow \text{unify } t_1 t_2 \sigma.$

439 We also assume that refreshing variables does not affect unification. In particular a *refresh* for a
 440 term t is a function from variables to variables that: (i) is defined on all the variables in t ; (ii) is
 441 injective.

442 ► **Definition 16** (`refresh_for x t`). $(\text{vars } t \subseteq \text{domf } x) \wedge \text{injective } x \wedge (\text{domf } x \# \text{codomf } x).$

443 ► **Axiom 4** (`unif_ren`). $\forall x y z w t_1 t_2,$
 $\text{refresh_for } w t_1 \rightarrow \text{refresh_for } y t_2 \rightarrow \text{refresh_for } z t_1 \rightarrow \text{refresh_for } x t_2 \rightarrow$
 $\text{codomf } w \# \text{vars } t_1 \cup \text{vars } (\text{rename } y t_2) \rightarrow$
 $\text{codomf } z \# \text{vars } t_1 \cup \text{vars } (\text{rename } x t_2) \rightarrow$
 $\text{unify } (\text{rename } w t_1) (\text{rename } y t_2) \varepsilon \rightarrow \text{unify } (\text{rename } z t_1) (\text{rename } x t_2) \varepsilon.$

444 In Figure 1, the rules of the program are equipped with three signatures, one per predicate.
 445 Besides the arity of each predicate, a signature specifies which arguments are inputs and which are
 446 outputs, their types, and whether the predicate is deterministic. They indicate that **f** and **g** are
 447 expected to be functions, as specified by the **func** keyword, whereas **r** is a relation, as indicated by
 448 the **pred** keyword. The $\rightarrow$ symbol separates inputs from outputs; therefore, all three predicates are
 449 expected to take an **int** as input and return an **int** as output.

450 The signatures of the program are stored in the **sig** field of the program and are encoded as
 451 described in [12]; that is, similarly to the code in Figure 1, we use arrows for term application, data
 452 types to encode base types such as *int*, and two dedicated symbols to distinguish deterministic from
 453 non-deterministic predicates.

454 A program execution behaves deterministically if, given a program, a substitution, and an
 455 input tree for the `runT` inductive, the execution either fails or, if it succeeds, the resulting tree of
 456 alternatives is $\square$.

457 ► **Definition 17** (`is_det p σ v t`). $\forall r, \text{runT}' p v \sigma t r \rightarrow r = \square \vee \exists \sigma, r = \cdot(\sigma, \square).$

458 The theorem we want to prove is the following one:

459 ► **Theorem 18** (`det_check_call`). $\forall p \sigma t v,$
 $\text{check}_\mathbb{P} p \rightarrow \text{tm_is_det } p.(\text{sig}) t \rightarrow \text{is_det } p \sigma v (\text{Todo } (\text{call } t)).$

460 Here, `checkℙ` denotes the function that checks the program with respect to mutual exclusion
 461 and rule checking, and `tm_is_det` denotes the function that tests whether the head of the term
 462 refers to a rigid constant with a deterministic type.

463 The proof of this lemma proceeds by induction on the program execution. However, without
 464 further work, the induction hypothesis is too weak since it talks about `Todo` trees (i.e. atoms),
 465 whereas we need it on any tree.

466 To address this issue, we introduce the notion of deterministic trees (`det_tree`), show that a tree
 467 satisfying this condition enjoys the desired properties, and then prove the following more general
 468 theorem:

469 ► **Lemma 19** (`det_check_tree`). $\forall \sigma v p t, \text{check}_\mathbb{P} p \rightarrow \text{det_tree } p.(\text{sig}) t \rightarrow \text{is_det } p \sigma v t.$

470 Our target theorem `det_check_call` is a corollary of `det_check_tree`.

471 7 Related work

472 We studied a PROLOG semantics in which input arguments are *matched*, rather than unified, against
 473 rule heads. This choice agrees with the ELPI interpreter, which is in turn inspired by B-Prolog's
 474 “matching-clauses” [1, 28]. The two semantics we presented (and proved equivalent) also apply to
 475 standard PROLOG if one forces all predicate signatures to have only output arguments.

For determinacy analysis, matching input arguments affects the proofs only as follows: Axiom 1 does require the query q to be ground. Adapting our results to standard PROLOG therefore requires the insertion of two additional hypotheses in Theorem 18. First, one has to require that t has ground inputs. Second, that the program p is well-moded. Well-moded predicates produce ground outputs when given ground inputs; hence, starting from an initial query with ground inputs, well-moded programs behave exactly as if input arguments were matched. Therefore, the proofs in Section 6 still apply.

The only mechanization of Prolog’s semantics we could find is the one by Pusch in ISABELLE/HOL [21] that is based on the model of Debray and Mishra [6, Sec. 4.3]. Their work starts from that semantics and refines it by introducing some optimizations usually found in the Warren abstract machine [27, 1]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in Section 4 that pushed us to develop a more explicit semantics (with respect to cut). In a sense, our work is complementary. By combining both works one can show the tree-based semantics is equivalent to an optimized stack-based one.

Another relevant work is by King [16], but we could not find their Rocq source code. Their semantics is denotational and cannot easily cover all the features that [12] covers, but it would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by [2], and we thank Y. Bertot for insightful discussions on the topic. In [2], Bertot mechanizes the correctness of a compiler. He relates the semantics of an imperative language to that of its compiled assembly code and proves their equivalence. His approach uses the compiler as the translation function from one language to the other, but does not introduce a decompiler, just as we do not define a function from stacks to trees. To show that the assembly code behaves like the imperative language, he proceeds by induction on the execution of the assembly program. We adopt the same strategy in the proof of Theorem 15.

8 Conclusion

This paper describes two operational semantics for Prolog with cut. The first semantics is based on And–Or trees: it is well suited to graphically illustrating the scope of cut, and to proving lemmas about the impact of its evaluation when reasoning about disjunctions. The second semantics is based on a stack of alternatives: it is computationally more efficient, but it loses the structural expressiveness of the tree-based semantics. Thanks to a dedicated function, we translate trees into stacks and prove the two semantics equivalent. This stack-based semantics corresponds to the first-order fragment of ELPI. Using the tree-based semantics, we mechanize the first-order part of the determinacy checker that has been available in ELPI since version 3.0, and we prove that a call to a deterministic predicate returns at most one solution.

The formal development consists of approximately 2,500 lines of specifications and about 5,000 lines of *ssreflect* proofs. Roughly 30% of the code is dedicated to the tree semantics (in particular, the proofs in Section 3.4), 15% to the stack semantics, and 35% to the equivalence proof. The remaining 20% is devoted to the determinacy checker. The development took approximately 8 months.

To fully model the ELPI language and mechanize [12], we would need to account for higher-order variables, i.e. variables representing predicates. This extension is ongoing and largely orthogonal to the present paper, since cut and higher-order features do not interact in subtle ways.

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